

Categories for the working bioinformatician: theoretical foundations for UMAP

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- Takes in a finite set of data points assumed to live on a Riemannian manifold, possibly of high dimension
- Using techniques that help to preserve both local and global information, reduces dimension of data from potentially very high numbers

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- Based on Spivak's work on a singular complex functor for (ep-)metric spaces, the authors are able to keep track of the global and local structure simultaneously
- The global structure, i.e. the topology, of the data is stored in the simplicial structure, just as in the case of a standard topological space
- However, incorporating fuzziness into the structure preserves more of the local, geometric information: the distance(s) encoded by the Riemannian metric(s) are mapped into the base interval for the sheaves comprising the fuzzy structure, packaging both the local and global information into one structure

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- Worth noting that the use of simplicial sets is not strictly necessary— given that the data is assumed to exist (approximately) on a Riemannian manifold, one could theoretically triangulate the data into the more commonly used structure of a simplicial complex

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- Worth noting that the use of simplicial sets is not strictly necessary— given that the data is assumed to exist (approximately) on a Riemannian manifold, one could theoretically triangulate the data into the more commonly used structure of a simplicial complex
- While the algorithm could certainly exist in its current state with simpler foundations such as these, such a choice would significantly limit the flexibility and potential for expansion of the work in the future

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- Their goal was to build the algorithm on a strong categorical foundation allowing for future expansion and improvement of the algorithm, along with other future work
- Category theory relies heavily on selecting for certain desirable properties (such as functoriality and naturality) in the structures used, making it a particularly robust framework for incorporating techniques from a wide range of mathematical disciplines

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- In contrast, simplicial complexes are only guaranteed finite limits and colimits, severely limiting such constructions
- Finally, simplicial sets generalize much more readily to simplicial objects in more general categories, potentially allowing for processing of more highly structured data, such as a Lie group structure (keeping track of symmetries in the data)

A whirlwind tour of category theory

1. Category theory

- Motivation and philosophy
- Definitions
- Familiar examples
- Diagrams, duality, and functors
- Universal constructions
- Adjunctions

2. Simplicial sets

3. Fuzzy mathematics

Motivation and philosophy

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- When possible, work with objects defined uniquely (up to isomorphism) by certain properties
- The principle of equivalence: isomorphic objects should be treated as the same

Heuristic/Graphical definition

- We can think of a category as a special version of a directed multigraph
- The vertices are known as *objects*, and the edges are known as *morphisms*
- Note: these collections can each be “larger” than a set (i.e. a proper class)
- There is a composition law: for each sequential pair of morphisms $a \xrightarrow{f} b \xrightarrow{g} c$ there is a unique arrow $a \xrightarrow{h} c$ such that $g \circ f = h$
- Each vertex/object c admits a unique loop, the *identity morphism*

Hom objects

- For each pair of objects $x, y \in \mathcal{C}$, denote the collection of all morphisms $x \rightarrow y$ by $\text{Hom}_{\mathcal{C}}(x, y)$
- If for each pair of objects in $\mathcal{C}$, this collection is a set, then $\mathcal{C}$ is known as a *locally small* category

Set-theoretic examples

- Set
 - Objects: sets
 - Morphisms: set functions

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- Any poset: $\mathbb{Z} \times \mathbb{Z}$
 - Objects: pairs $(x, y) \in \mathbb{Z} \times \mathbb{Z}$
 - Morphisms: $(x_1, y_1) \rightarrow (x_2, y_2)$ if and only if $x_1 \leq x_2$ and $y_1 \leq y_2$

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- Any totally-ordered set: $\mathbb{R}$
 - Objects: real numbers $r \in \mathbb{R}$
 - Morphisms: $x \rightarrow y$ if and only if $x < y$

Variations on spaces

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 - Objects: simplicial complexes
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- **Ho(Top)**
 - Objects: topological spaces
 - Morphisms: homotopy classes of maps
- **Loc**
 - Objects: Locales; i.e. topological spaces without points
 - Morphisms: continuous maps of locales (i.e. maps for which preimages of opens are open)

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- $\mathbf{Ab}$: the full subcategory of $\mathbf{Grp}$ on the abelian groups
- $\mathbf{Ch}_\bullet(\mathbf{Ab})$
 - Objects: Chain complexes of abelian groups
 - Morphisms: Chain maps, i.e. sequences of morphisms of abelian groups that commute with the differentials

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- Preserves identities and isomorphisms

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Let $\mathcal{C}$ be a category. Define a new category $\mathcal{C}^{\text{op}}$ with the same objects as $\mathcal{C}$, but with all of the arrows reversed. We call $\mathcal{C}^{\text{op}}$ the *opposite category* of $\mathcal{C}$.

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Definition

$\text{CAlg}_{\mathbb{R}}$ (*commutative $\mathbb{R}$ -algebras*):

- Objects: vector spaces V over $\mathbb{R}$ with a commutative, unital, and distributive multiplication operation
- Morphisms: linear maps respecting the multiplication operation

Definition

$\text{Aff}_{\mathbb{R}}$ (*affine schemes over $\mathbb{R}$*):

- Objects: $\text{Spec } V := \text{Hom}_{\text{CAlg}_{\mathbb{R}}}(-, V) : \text{CAlg}_{\mathbb{R}} \rightarrow \text{Set}$ for each $V \in \text{CAlg}_{\mathbb{R}}$
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- Basis for the modern approach to (real) algebraic geometry

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- Objects come with graded abelian groups dual to homology known as *cohomology*
- Can build a cochain complex out of a space dual to our usual singular chain complex, yielding the *singular cohomology* of the space
- Not quite as easy to describe geometrically as homology, but often has more structure and is a stronger invariant of the space

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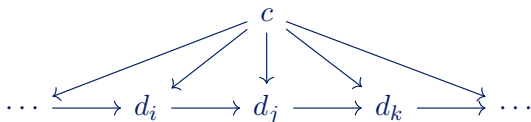
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- In **Grp** and **Ab**, the trivial group $\{0\}$ is both initial and final

Cones over diagrams

Definition

Given a diagram $D \in \text{Fun}(I, \mathcal{C})$, a *cone* over D is an object $c \in \mathcal{C}$ equipped with a unique morphism to each object of D . A morphism of cones over D is given by a morphism in $\mathcal{C}$ that commutes with all of the morphisms in D .



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- Almost any other universal construction can be defined as the (co)limit of some diagram

Familiar universal constructions

- Quotients of any category where you can define equivalence relations can be constructed as a colimit of a certain diagram
- Intersections of topological spaces and sets are an example of a limit known as a *pullback*
- Cartesian products of sets are an example of a limit known as a *product*

Discrete foundations for homotopy theory

1. Category theory

2. Simplicial sets

- The simplex category
- Definition and examples
- The topos structure
- The $|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : \mathbf{Sing}(-)$ adjunction

3. Fuzzy mathematics

The simplex category Δ

- Let Δ be the category with objects finite totally ordered sets of the form

$$[n] = \{0 \rightarrow 1 \rightarrow \cdots \rightarrow n\}$$

- Morphisms are order-preserving maps of finite sets subject to a list of identities known as the *simplicial identities*
- Can think of the set $[n]$ as an abstract version of the “standard n -simplex”
- We call Δ the *simplex category*

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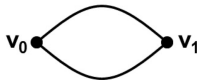
- Each face map sends an i -simplex to one of its $i - 1$ -dimensional faces; each degeneracy inserts an i -simplex as a degenerate $i + 1$ -simplex

Comparison to complexes

- Simplicial sets allow for “worse” geometric scenarios in order to model non-triangulable spaces

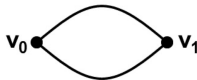
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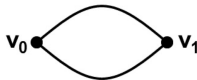
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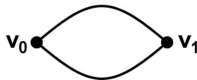
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- All of the simplices are assigned a total ordering, allowing each face and degeneracy map to be defined uniquely
- Simplicial maps are entirely determined by their values on the non-degenerate simplices

The Quillen adjunction

Theorem (Quillen)

There are functors $|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : \mathbf{Sing}(-)$, known as geometric realization and singular complex, extending those defined for simplicial complexes, that form an adjunction

- An *adjunction* is a pair of functors between two categories that induce isomorphisms between Hom objects in the two categories
- Can think of an adjunction as a pair of functors that allow one to move between two categories in a convenient manner

Repurposing a controversial field of logic

1. Category theory

2. Simplicial sets

3. Fuzzy mathematics

- ep-metric spaces
- Fuzzy simplicial sets
- Spivak's adjunction: the heart of UMAP

Extended pseudo-metric spaces

- An *extended pseudo-metric space* is a generalization of a metric space that allows for infinite distances and a slight weakening of the triangle inequality
- Finite sets of points on a Riemannian manifold form examples of finite ep-metric spaces
- Such data sets are the input for the UMAP algorithm

Fuzzy simplicial sets

- A *fuzzy simplicial set* is a functor $[0, 1] \rightarrow \mathbf{sSet}$ with the property that it “varies continuously” along the interval in a precise manner
- More specifically, we can think of such an object as a simplicial set where each simplex is assigned a weight in the interval $[0, 1]$
- The weight of a simplex will be used to track the distances in an ep-metric space, while the simplicial structure tracks the topological structure

Spivak's adjunction

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